Ninth Grade Tile Task

Standard A.APR.1

9

1. a) Coefficient of x in $(x + 3)(x + 0)$ **3**

b) Coefficient of x in $(x + 4)(x + 2)$ **6**

c) Constant term of $(x + 3)(x + 0)$ **0**

2. a) Constant term of $(x + 1)(x + 1)$ **1**

b) Constant term of $(x + 3)(x + 3)$ **9**

c) Constant term of $(x + 2)(x + 1)$ **2**

3. a) Coefficient of x in $(x + 3)(x + 5)$ **8**

b) Constant term of $(x + 5)(x + 1)$ **5**

c) Coefficient of x in $(x + 1)(x + 3)$ **4**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Ninth Grade Tile Task

Standard A.CED.1

9

1. a) Solve 4x + 4 = 24: x = ? **5**

b) Solve 3x + 6 = 6: x = ? **0**

c) Solve 5x + 6 = 21: x = ? **3**

2. a) Solve 8x + 1 = 9: x = ? **1**

b) Solve 1x + 5 = 14: x = ? **9**

c) Solve 2x + 0 = 4: x = ? **2**

3. a) Solve 6x + 4 = 28: x = ? **4**

b) Solve 1x + 0 = 8: x = ? **8**

c) Solve 7x + 8 = 57: x = ? **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: NINTH GRADE

Ninth Grade Tile Task

Standard A.CED.2

9

1. a) Slope of the line through (0, 0) and (1, 1) — 1

b) Slope of the line through (0, 2) and (1, 5) — 3

c) Slope of the line through (0, 4) and (1, 12) — 8

2. a) Slope of the line through (0, 0) and (1, 4) — 4

b) Slope of the line through (0, 3) and (1, 8) — 5

c) Slope of the line through (0, 3) and (1, 10) — 7

3. a) Slope of the line through (0, 3) and (1, 12) — 9

b) Slope of the line through (0, 1) and (1, 1) — 0

c) Slope of the line through (0, 1) and (1, 3) — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.CED.3

9

1. a) Solve x + y = 6, x - y = 6: x = ? **6**

b) Solve x + y = 5, x - y = 5: x = ? **5**

c) Solve x + y = 3, x - y = 3: x = ? **3**

2. a) Solve x + y = 2, x - y = 0: x = ? **1**

b) Solve x + y = 10, x - y = 6: x = ? **8**

c) Solve x + y = 2, x - y = 2: x = ? **2**

3. a) Solve x + y = 9, x - y = 5: x = ? **7**

b) Solve x + y = 18, x - y = 0: x = ? **9**

c) Solve x + y = 8, x - y = 0: x = ? **4**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: NINTH GRADE

Ninth Grade Tile Task

Standard A.CED.4

9

1. a) Solve 2y = 0: y = ? **0**

b) Solve 2y = 8: y = ? **4**

c) Solve 2y = 18: y = ? **9**

2. a) Solve 2y = 10: y = ? **5**

b) Solve 2y = 6: y = ? **3**

c) Solve 2y = 14: y = ? **7**

3. a) Solve 2y = 4: y = ? **2**

b) Solve 2y = 2: y = ? **1**

c) Solve 2y = 12: y = ? **6**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.REI.1

9

1. a) Solve 1x = 8: x = ? 8

b) Solve 7x = 7: x = ? 1

c) Solve 1x = 5: x = ? 5

2. a) Solve 2x = 8: x = ? 4

b) Solve 1x = 6: x = ? 6

c) Solve 2x = 6: x = ? 3

3. a) Solve 1x = 7: x = ? 7

b) Solve 1x = 2: x = ? 2

c) Solve 1x = 9: x = ? 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.REI.3

9

1. a) Largest whole number x with x < 7? **6**

b) Largest whole number x with x < 10? **9**

c) Largest whole number x with x < 4? **3**

2. a) Largest whole number x with x < 5? **4**

b) Largest whole number x with x < 2? **1**

c) Largest whole number x with x < 3? **2**

3. a) Largest whole number x with x < 1? **0**

b) Largest whole number x with x < 8? **7**

c) Largest whole number x with x < 6? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.REI.4

9

1. **a)** Positive solution of $x^2 = 25$ — 5

   b) Positive solution of $x^2 = 49$ — 7

   c) Positive solution of $x^2 = 1$ — 1

2. **a)** Positive solution of $x^2 = 9$ — 3

   b) Positive solution of $x^2 = 16$ — 4

   c) Positive solution of $x^2 = 64$ — 8

3. **a)** Positive solution of $x^2 = 36$ — 6

   b) Positive solution of $x^2 = 4$ — 2

   c) Positive solution of $x^2 = 81$ — 9

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.REI.5

9

1. a) Solve x + y = 9, x = 0: y = ? 9

b) Solve x + y = 9, x = 3: y = ? 6

c) Solve x + y = 7, x = 7: y = ? 0

2. a) Solve x + y = 8, x = 0: y = ? 8

b) Solve x + y = 5, x = 2: y = ? 3

c) Solve x + y = 8, x = 1: y = ? 7

3. a) Solve x + y = 9, x = 4: y = ? 5

b) Solve x + y = 4, x = 2: y = ? 2

c) Solve x + y = 5, x = 4: y = ? 1

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.REI.6

9

1. a) Solve x + y = 18 and x - y = 0: y = ? **9**

b) Solve x + y = 16 and x - y = 2: y = ? **7**

c) Solve x + y = 8 and x - y = 0: y = ? **4**

2. a) Solve x + y = 3 and x - y = 1: y = ? **1**

b) Solve x + y = 4 and x - y = 0: y = ? **2**

c) Solve x + y = 17 and x - y = 1: y = ? **8**

3. a) Solve x + y = 11 and x - y = 5: y = ? **3**

b) Solve x + y = 14 and x - y = 4: y = ? **5**

c) Solve x + y = 9 and x - y = 9: y = ? **0**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard A.SSE.1

9

1. a) Coefficient of x in $5x^2 + 1x$ — **1**

b) Coefficient of x in $4x^2 + 0x$ — **0**

c) Coefficient of x in $6x^2 + 9x$ — **9**

2. a) Coefficient of x in $2x^2 + 3x$ — **3**

b) Coefficient of x in $4x^2 + 5x$ — **5**

c) Coefficient of x in $3x^2 + 2x$ — **2**

3. a) Coefficient of x in $5x^2 + 7x$ — **7**

b) Coefficient of x in $8x^2 + 4x$ — **4**

c) Coefficient of x in $2x^2 + 6x$ — **6**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Ninth Grade Tile Task

Standard A.SSE.2

9

1. a) $x^2 + 6x + 0 = (x + 0)(x + ?)$. Build ? **6**

b) $x^2 + 6x + 5 = (x + 1)(x + ?)$. Build ? **5**

c) $x^2 + 8x + 7 = (x + 1)(x + ?)$. Build ? **7**

2. a) $x^2 + 9x + 0 = (x + 0)(x + ?)$. Build ? **9**

b) $x^2 + 4x + 4 = (x + 2)(x + ?)$. Build ? **2**

c) $x^2 + 3x + 2 = (x + 2)(x + ?)$. Build ? **1**

3. a) $x^2 + 0x + 0 = (x + 0)(x + ?)$. Build ? **0**

b) $x^2 + 8x + 0 = (x + 0)(x + ?)$. Build ? **8**

c) $x^2 + 5x + 4 = (x + 1)(x + ?)$. Build ? **4**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: NINTH GRADE

Ninth Grade Tile Task

Standard F.BF.1

9

1. a) $f(x) = x + 7, g(x) = x. f(g(1)) = ?$ **8**

b) $f(x) = x + 0, g(x) = x. f(g(0)) = ?$ **0**

c) $f(x) = x + 5, g(x) = x. f(g(2)) = ?$ **7**

2. a) $f(x) = x + 2, g(x) = x. f(g(0)) = ?$ **2**

b) $f(x) = x + 2, g(x) = x. f(g(2)) = ?$ **4**

c) $f(x) = x + 6, g(x) = x. f(g(0)) = ?$ **6**

3. a) $f(x) = x + 0, g(x) = x. f(g(1)) = ?$ **1**

b) $f(x) = x + 1, g(x) = x. f(g(2)) = ?$ **3**

c) $f(x) = x + 5, g(x) = x. f(g(4)) = ?$ **9**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: NINTH GRADE

Ninth Grade Tile Task

Standard F.IF.1

9

1. a) f(x) = 3x + 1. Find f(2). **7**

b) f(x) = 1x + 7. Find f(2). **9**

c) f(x) = 1x + 4. Find f(4). **8**

2. a) f(x) = 2x + 1. Find f(0). **1**

b) f(x) = 2x + 2. Find f(0). **2**

c) f(x) = 1x + 0. Find f(3). **3**

3. a) f(x) = 1x + 4. Find f(1). **5**

b) f(x) = 2x + 0. Find f(0). **0**

c) f(x) = 2x + 0. Find f(2). **4**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

Ninth Grade Tile Task

Standard F.IF.2

9

1. a) $f(x) = x + 1$. Find $f(1)$. 2

b) $f(x) = x + 0$. Find $f(4)$. 4

c) $f(x) = x + 7$. Find $f(2)$. 9

2. a) $f(x) = x + 2$. Find $f(1)$. 3

b) $f(x) = x + 5$. Find $f(1)$. 6

c) $f(x) = x + 1$. Find $f(0)$. 1

3. a) $f(x) = x^2$. Find $f(0)$. 0

b) $f(x) = x + 5$. Find $f(2)$. 7

c) $f(x) = x + 8$. Find $f(0)$. 8

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard F.IF.4

9

1. a) y-intercept of y = 0x + 2 **2**

b) y-intercept of y = 2x + 6 **6**

c) y-intercept of y = 1x + 7 **7**

2. a) y-intercept of y = 0x + 1 **1**

b) y-intercept of y = 0x + 0 **0**

c) y-intercept of y = 2x + 4 **4**

3. a) y-intercept of y = 2x + 3 **3**

b) y-intercept of y = 2x + 5 **5**

c) y-intercept of y = 1x + 8 **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard F.LE.1

9

1. a) Linear y = 2x + 1 at x = 1 — 3

b) Linear y = 6x + 1 at x = 1 — 7

c) Linear y = 8x + 1 at x = 1 — 9

2. a) Exponential y = 2^x: value at x = 1 — 2

b) Linear y = 5x + 1 at x = 1 — 6

c) Exponential y = 2^x: value at x = 3 — 8

3. a) Linear y = 4x + 1 at x = 1 — 5

b) Linear y = 0x + 1 at x = 1 — 1

c) Exponential y = 2^x: value at x = 2 — 4

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: NINTH GRADE

Ninth Grade Tile Task

Standard F.LE.2

9

1. **a)** Rate of change of the line through (0, 3) and (1, 3) — 0
   b) Rate of change of the line through (0, 4) and (1, 13) — 9
   c) Rate of change of the line through (0, 2) and (1, 7) — 5

2. **a)** Rate of change of the line through (0, 1) and (1, 8) — 7
   b) Rate of change of the line through (0, 1) and (1, 5) — 4
   c) Rate of change of the line through (0, 2) and (1, 10) — 8

3. **a)** Rate of change of the line through (0, 1) and (1, 2) — 1
   b) Rate of change of the line through (0, 0) and (1, 2) — 2
   c) Rate of change of the line through (0, 2) and (1, 8) — 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard S.ID.1

9

1. **a)** A dot plot has 4 dots. How many data values? 4

   b) A dot plot has 5 dots. How many data values? 5

   c) A dot plot has 8 dots. How many data values? 8

2. **a)** A dot plot has 3 dots. How many data values? 3

   b) A dot plot has 7 dots. How many data values? 7

   c) A dot plot has 9 dots. How many data values? 9

3. **a)** A dot plot has 2 dots. How many data values? 2

   b) A dot plot has 1 dots. How many data values? 1

   c) A dot plot has 6 dots. How many data values? 6

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Ninth Grade Tile Task

Standard S.ID.2

9

1. a) Range of data values 5 and 0 — **5**

b) Range of data values 0 and 6 — **6**

c) Range of data values 3 and 7 — **4**

2. a) Range of data values 0 and 8 — **8**

b) Range of data values 4 and 4 — **0**

c) Range of data values 0 and 9 — **9**

3. a) Range of data values 5 and 4 — **1**

b) Range of data values 6 and 3 — **3**

c) Range of data values 5 and 3 — **2**

Tile box: cut out the tiles and move them to your answer boxes

0	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**

TILE TASKS: NINTH GRADE